

FizzBuzz Project

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README

User Documentation

What is this?

This program implements the classic FizzBuzz problem. To use it, run the `fizzbuzz.py` script. The function lists numbers from 1 to the number you provide, but for multiples of 3, it prints "Fizz" instead of the number, for multiples of 5, it prints "Buzz", and for multiples of both 3 and 5, it prints "FizzBuzz".

How do I use it?

1. [Install dependencies](#) if you do not already have them.
2. [Use the guide for the FizzBuzz function](#)

Installing Dependencies

- You need Python 3.x installed. To install Python, visit:
<https://www.python.org/downloads/>
- If you want to run tests, you need to install `pytest` :

```
pip install pytest
```

Using the fizzbuzz Function in Your Code

You can import the `fizzbuzz` function from `fizzbuzz.py` and use it in your own Python scripts or in the Python interactive shell.

Example Usage

1. Open a Python shell or create a new Python script in the same directory.
2. Import the function:

```
from fizzbuzz import fizzbuzz
```
3. Call the function with your desired range (e.g., 1 to 100):

```
fizzbuzz(100)
```

This will print the FizzBuzz sequence from 1 to 100.

Technical Documentation

- **fizzbuzz.py**: Contains the main logic for the FizzBuzz problem.
- **test_fizzbuzz.py**: Contains tests for the FizzBuzz implementation using pytest.
- **.github/workflows/fizzbuzz-tests.yml**: GitHub Actions workflow file that automatically runs the test suite on every push and pull request. It sets up a Python environment, installs dependencies, and runs pytest to ensure code quality and correctness.

The main function iterates from 1 to the number provided to the function and prints the appropriate output based on divisibility by 3 and/or 5.

Developer Documentation

Project Structure

- `fizzbuzz.py` : Main script.
- `test_fizzbuzz.py` : Test script.
- `.github/workflows/fizzbuzz-tests.yml` : Continuous Integration (CI) workflow for automated testing with GitHub Actions.

Continuous Integration

All tests are automatically run on GitHub Actions for every push and pull request using the workflow defined in `.github/workflows/fizzbuzz-tests.yml`. This ensures that all code changes are tested before being merged.

How to Run Tests Locally

To run the tests locally, make sure you are in the project root directory. Run the following command:

```
pytest
```

Pytest will automatically discover and run all test files named `test_*.py`.

Contribution Guidelines

- Follow PEP8 style guidelines.
- Write tests for new features or bug fixes.
- Document your code with comments where necessary.

Code for fizzbuzz function

Below you will find the code for the fizzbuzz function.

```
In [ ]: # fizzbuzz.py

# create a function for fizzbuzz
```

```
#define the function
def fizzbuzz(num_entry):

    # create checks to ensure that num_entry is valid

    #check if num_entry is numeric or integer
    # if num_entry isn't numeric
    if not isinstance(num_entry, int):
        # then give this error message
        raise TypeError("num_entry must be an integer")

    # if num_entry is 0

    if num_entry == 0 :
        # then give this error msg
        raise ValueError ("num_entry must be bigger than 0")

    # if num_entry is negative

    if num_entry < 0:
        raise ValueError("num_entry must not be negative")

    #start an empty list

    results = []

    # start a for loop for the fizzbuzz problem

    for x in range ( 1, num_entry+1):

        # if x is a multiple of 3 AND 5 then

        if x % 3 == 0 and x % 5 == 0:
            #append fizzbuzz to the list
            results.append("fizzbuzz")
        # if the x is a multiple of 3 then
        elif x % 3 == 0:
            #append fizz to the list
            results.append("fizz")

        # if the x is a multiple of 5

        elif x % 5 == 0:
            # append buzz to the list
            results.append("buzz")

        # if none of those conditions are met then

        else:
            results.append(x)

        # return the result list

    return results
```

GitHub Actions Workflow (YAML)

Below is the code for the .yml file used for the GitHub actions workflow for testing the function.

```
name: Fizzbuzz tests

# specify when we want this to be triggered
# we want to trigger it for every push and PR

on: [push, pull_request]

#specify the jobs needed for the CI

jobs:
  tests:
    runs-on: ubuntu-latest

    steps:

      - uses: actions/checkout@v4

      - name: Set up python
        uses: actions/setup-python@v5
        with:
          python-version: "3.x" # to use any version of python 3

      #make sure we have pytest and upgrades if needed
      - name: Install dependencies
        run: pip install --upgrade pip pytest
      # run the actual test
      - name: Run pytest
        run: pytest
```